

## 1. What do you understand by Database?

A database is an organized collection of data that is stored, managed, and retrieved electronically. It allows for efficient data manipulation, storage, and retrieval. Databases are used in various applications such as banking, e-commerce, education, and healthcare, to manage large volumes of information systematically and this is all database.

## 2 . What is Normalization?

Normalization is the process of organizing data in a database to minimize redundancy and dependency by dividing the data into multiple tables and defining relationships between them. The goal is to:

- Eliminate redundant data.
- Ensure data integrity and consistency.
- Simplify maintenance

## 3. What is the Difference between DBMS and RDBMS?

### **DBMS**

**Definition** Database Management System manages data without relations between tables

**Structure** Stores data in files

**Normalization** May not support normalization

**ACID Properties.** May not strictly enforce ACID properties.

### **RDBMS**

**Relational Database Management System** manages data with relationships between tables

**Stores data in tables with rows and columns.**

**Supports normalization to reduce redundancy**

**Ensures ACID compliance for transactions**

## 4. What is E.F. Codd's Rule of RDBMS Systems?

E.F. Codd proposed **12 rules** to define a fully functional RDBMS. Some key rules include:

1. **Information Rule:** All information is stored in tables.
2. **Guaranteed Access Rule:** Every piece of data is accessible by table name, primary key, and column name.
3. **Systematic Treatment of Null Values:** Nulls must be uniformly treated.
4. **Dynamic Online Catalog:** Metadata should also be stored in tables and accessible through SQL.
5. **Comprehensive Data Sub-language Rule:** The system must support a single language for data definition, manipulation, and authorization.

## 5. What do you understand by Data Redundancy?

- Increased storage costs.
- Difficulty in maintaining consistency.
- Risk of conflicting data during updates.

## 6. What is a DDL Interpreter?

- Interpreting and executing SQL commands related to database schema, such as creating, altering, or dropping tables and indexes.
- Translating these commands into actions that the database management system can perform

## 7. What is a DML Compiler in SQL?

The DML (Data Manipulation Language) Compiler translates DML statements (like select, insert, update, delete) into low-level instructions that the database engine can execute. It optimizes the query for better performance and retrieves or modifies the data as requested.

## 8. What is SQL Key Constraints?

```
CREATE TABLE Students (
```

```
    StudentID INT PRIMARY KEY,
```

```
    Name VARCHAR(50) NOT NULL,
```

```
    Age INT CHECK (Age >= 18),
```

```
    ClassID INT,
```

```
FOREIGN KEY (ClassID) REFERENCES Classes(ClassID)  
);
```

### **9. What is a Savepoint? How to Create a Savepoint?**

```
BEGIN TRANSACTION;  
  
UPDATE Employees SET Salary = Salary + 1000 WHERE Department = 'HR';  
  
SAVEPOINT SavePoint1;  
  
UPDATE Employees SET Salary = Salary + 2000 WHERE Department = 'Finance';  
  
ROLLBACK TO SavePoint1;  
  
COMMIT;
```

### **10. What is a Trigger and How to Create a Trigger in SQL?**

```
CREATE TRIGGER AfterInsertEmployee  
  
AFTER INSERT ON Employees  
  
FOR EACH ROW  
  
BEGIN  
  
    INSERT INTO AuditLog (Action, Timestamp)  
  
    VALUES ('New employee added', NOW());  
  
END;
```